# REMEMBERING: A STUDY IN EXPERIMENTAL AND SOCIAL PSYCHOLOGY.

By F. C. BARTLETT, M.A., F.R.S. (Cambridge: at the University Press, 1932. Pp. vii+317. 16s.)

PROFESSOR BARTLETT'S volume is by no means to be regarded as just another monograph on memory. In more or less detail, it covers almost the whole of the cognitive side of mental life. Starting from memory as an activity midway up the scale, he shows that the processes of remembering are not sharply different from the lower processes of sense-perception or from the higher processes of constructive imagination and thinking. In fact, the theory of clear-cut mental levels comes to look almost as misleading as the doctrine of clear-cut mental faculties. Further, although he deals primarily with intellectual processes, Professor Bartlett is no mere intellectualist. Believing that psychology is a biological science, he adopts throughout a functional rather than a merely descriptive standpoint. His book is not a work on "memory" but on "remembering." He is not concerned to analyze or to catalogue what we recollect or what we perceive; his interest centres rather upon the conditions, upon the wide variety of conditions, under which recollecting and perceiving actually take place. Thus he continually insists upon the influence of non-intellectual or affective factors—on the interplay of appetites, instincts, interests, and ideals.

This unifying outlook, whereby all the details of our mental life are viewed as part of one consistent interacting whole, forms the most remarkable and the most suggestive feature of Professor Bartlett's work. As a result, his volume is by far the most important contribution to psychology that has appeared in this country during recent years.

Professor Bartlett relates the history of his experimental work. One bright May afternoon in the year before the war, the Psychological Laboratory at Cambridge was formally opened. As part of the demonstrations, Dr. Bartlett was invited to show in a darkened room a number of pictures, geometrical shapes and optical illusions to a long queue of inquisitive visitors. The remarks and interpretations were so various and intriguing that he thereupon commenced a series of systematic experiments proceeding along similar lines. He abandoned the "exact

methods of nonsense-syllables," which were introduced by Ebbinghaus and which have dominated work on memory ever since ; and, adopting more natural and realistic material, tried to bring his study into closer relation with remembering in every-day life. This enables him to get away from the rather abstract and artificial conditions that so commonly obtain in the ordinary laboratory experiment, and brings his procedure nearer to that of the clinician.

He begins with experiments on perception. He shows his subjects, for a fraction of a second, various kinds of visible material. In the first experiments it consists of simple patterns and designs, made up perhaps of three or four lines only ; in the second, a set of progressive patterns—patterns in which, throughout a series of successive views, the subject sees, as it were, one and the same design being progressively built up. He then proceeds to more concrete representations. And finally ends with reproductions of well-known pictures.

What are the results ? With most observers (the trained psychologists seem often to form exceptions) the first reaction, on getting a glimpse of the material, is to treat whatever they see as a single unitary whole, with perhaps one or two features standing out in dominance. As a rule, the ordinary person does not take the picture or pattern item by item and methodically build up the whole ; the first thing to emerge is an " attitude," often a feeling of surprise, familiarity, or dislike. He has an overmastering tendency to jump to a general impression, and on the basis of this guesses at the probable detail. His construction seeks, as it were, to justify his general impression, to satisfy or fortify his general attitude. As a result, the interpretation may at times go wildly astray. For example, after seeing a slide of Yeames' picture of *Hubert and Little Arthur*, one observer declared it showed " a woman in a white apron with a child by her knee." Another said it was " a representation of Othello saying to Desdemona ' Come fly with me.' " A third suggested " It might very well be ' The Woman taken in Adultery ' " ; subsequently, however, he thought the slide depicted Charles I and Henrietta.

The non-psychological are apt to suppose that, of all our cognitive reactions, the process of perception is one of the most simple and immediate, as it certainly is one of the most fundamental ; yet these experiments reveal what modern psychology has long suspected, namely, that the process of perception is exceedingly complex. Indeed, perception appears at once to be the starting point of what are more commonly regarded as processes of memory.

Professor Bartlett's second set of researches deals with imaging. Here his experiments are based on the familiar device of looking at smudged ink blots, and seeing what can be made of them, much as one sometimes finds shapes in clouds or sees faces in the fire. The interpretations show a strong preoccupation with animals or human beings; but the most striking feature of all is their enormous variety. A blot which to one looks like a tortoise may seem to another to be two dead ducks and an ostrich; a third will declare that it is a dog worrying a table-cloth; and a fourth suggests a picture of Sohrab and Rustum from Arnold's well-known poem. The outcome is to throw into strong relief a process which Professor Bartlett points out occurs and recurs in every cognitive reaction—in perceiving, in imagining, in remembering, and in reasoning—namely, the effort after meaning.

The experiments on remembering itself are more extensive. In the first the subject is shown a series of picture postcards representing the face of a soldier or a sailor of varying rank: (it will be recalled that the experiments were started during the early days of the war when there was a widespread interest in the fighting services). After an interval, the subject describes the various cards in order, and answers questions about the details. At once it appears that accurate recall is the exception and not the rule. So long as psychologists worked primarily with nonsense-syllables, the very material employed tended to throw excessive stress on mere reproduction. Substitute material more akin to what we seek to remember in ordinary every-day life, and literal reproduction becomes as rare as it is unnecessary. The subject himself may be unaware of the changes and distortions. For example, the emergence of a memory-image may positively mislead the subject's recollection, while at the same time increasing his self-confidence: here, as in the case of actual perception, seeing, even with the mind's eye, appears to be believing. The main conclusion drawn is that "remembering appears to be far more an affair of construction than of reproduction."

In the second set of experiments, the subject is given a story or a drawing, and is subsequently required to reproduce it again and again at intervals of increasing length. The results are strongly suggestive of the way in which a rumour may arise on the basis of a simple fact, get gradually distorted, and then settle down in the shape of a fixed traditional legend. A further set of experiments employs the ingenious method of picture writing. The subjects learned to connect a series of hieroglyphs with the words they were supposed to represent; they were then to write from dictation a short story, inserting wherever they could, not words but the hieroglyphic signs. A number of instruc-

tive inferences are drawn, some of them quite unexpected. For example, a special determination to remember was often promptly followed by complete omission.

In the last set of experiments, a picture, a story, or an argument is reproduced not by one person but by a series. The reproduction of the first is handed on to the second; the reproduction of the second is passed on to the next; and so on progressively, until at length the final version, abridged, conventionalized, and variously transformed by a cumulative sum of minor modifications, emerges almost unrecognizable. The procedure will no doubt be familiar to many from the parlour game which is variously known as "Gossip" or "Russian Scandal." The process as a whole shows many similarities to the changes of popular stories as they pass from mouth to mouth, and a few striking differences. It is found that the cumulative recall of a very few people may result in the production of a totally new event or story. Chains of reasoning may entirely disappear. The final opinion may be exactly the reverse of the original statement from which it is derived.

The results of all these experiments are gathered together in a series of theoretical chapters. Professor Bartlett starts with a world of human beings confronted by an environment in which they can survive and succeed only so far as they can match its infinite diversity by an increasing variety of response and at the same time discover ways of escaping from the domination of what is actually present. Past experience is commonly said to operate by producing some change in the nervous system. Traces, after-effects, are somehow left upon the cortex of the brain. Professor Bartlett insists that this simple explanation is inadequate; for it is usually implied that the traces so left behind are inert, inactive, and individual, without any organization of their own. The re-excitement of brain-traces might suffice to explain the reproduction of nonsense words or of fixed mechanical habits. But in actual life what seems to operate and persist is rather some active, developing pattern, some form of arrangement, which is not rigid, self-contained, and unchanging, but is itself a living organization of past reactions and experiences. Borrowing a term from Sir Henry Head, Professor Bartlett speaks of these organized patterns as schemata or schemes.

In a simple creature, real or pre-supposed for text-book purposes, the fixity of habit tends to suggest that the past affects the present by forcing the creature to repeat old sequences, in a stereotyped chronological order which cannot readily be broken. For the higher and more complex organism, therefore, the problem is this: how to resolve the

original scheme into its elements, and so transcend the original order in which those elements occurred. This problem, according to Professor Bartlett's view, can be solved solely by the aid of consciousness. It is a problem which gives to consciousness its pre-eminent function. Through consciousness the organism is enabled " to turn round upon its own schemata and make them the objects of its reactions."

To overcome the difficulty, the method of images is first evolved. " Images are a device for picking bits out of schemes, for increasing the chance of variability in the re-construction of past stimuli and situations, for surmounting the chronology of presentations." Thus, in any process of recall, images (as the experiments reveal) are particularly liable to arise when any slight check occurs : the typical case for their occurrence is " the arousal of cross-streams of interest."

The device of images, however, possesses several deficiencies. Of these, two are singled out as especially important. The image, and in particular the visual image, " is apt to go farther in the direction of the individualization of situations than is biologically useful " ; and the principles of the combination of images have their own peculiarities and " result in constructions which are relatively wild, jerky, and irregular, compared with the straightforward unwinding of a habit, or with the somewhat orderly march of thought." As Professor Bartlett rightly points out in passing, the familiar forms of association—the principles of contiguity, succession, and similarity, a logical classification mainly derived from the analysis of word-associations—are not sufficient to describe the peculiar modes in which images combine. These latter modes seem at bottom to be dependent on the special conditions that determine the combination, not of words or of ideas that are verbally expressed, but rather of interests and attitudes.

The defects of the image, therefore, are met by the later device of words. Words no doubt arise under social pressure. They are explicitly analytic—far more analytic than images. They can be used not only to describe particular features but also to indicate the qualitative and the relational factors in the general aspect. Hence to some extent they tend to compensate for the extreme particularity of the image ; and thus, in Professor Bartlett's experiments, they often appear as an alternative form of recall. They are, in fact, the best of all human inventions for perfecting reaction at a distance, and are essential to all the higher processes of thought.

The second part of Professor Bartlett's book deals with remembering as a study of social psychology. Here he discusses the process of con-

ventionalization and Jung's notion of a collective unconsciousness ; and so adds to his views on individual memory an examination of the basis of social recall. This portion of the work embodies some first-hand observations upon social recall among the natives of Swaziland ; but it would be impossible in a single short review to do justice to the suggestive views put forward or to the penetrating things said by the way. It must be sufficient to add that here again Professor Bartlett has struck upon a field and a method of investigation which have hitherto been neglected by the psychologist, but which, as he clearly demonstrates, should yield results of theoretical value and possibly of great practical importance.

CYRIL BURT.

ERRATA IN ARTICLE ON
INTERESTS AND MOTIVES FOR STUDY AMONG ADULT
EVENING STUDENTS.

VOLUME III, PART 1.

Page 15. Table for question 9—for 15% read 12%.
Page 15. Table for question 10—for 34% read 36%.
Page 15. Table for question 11—for 82% read 88%.
Page 16. Line 6, for 82% read 88%.